

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Dr.V.Parthipan

Code of Conduct:

I Jeevan Raaja S (192421005)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Computer Networks

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Short Answer Test – Questions with Answers

Assumption: 1 MB = 1,000,000 bytes, 1 GB = 1,000 MB, and 1 byte = 8 bits.

1. File Transfer using Transport and Data Link Layers Question

A company needs to transfer a **150 MB** design file over a **50 Mbps** network. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer adds **40 bytes** of header and trailer to each frame.

Calculate:

- Number of segments and frames
- Total Data Link overhead
- Transmission time
- Explain reliability.

Answer

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Given:

- File size = 150 MB = 150,000,000 bytes
- Segment size = 1500 bytes
- Overhead/frame = 40 bytes
- Bandwidth = 50 Mbps

Step 1: Number of Segments

$$\frac{150,000,000}{1500} = 100,000 \quad \frac{150,000,000}{1500} = 100,000$$

Segments = 100,000

Frames = Segments = 100,000

Step 2: Total Overhead

$100,000 \times 40 = 4,000,000$ bytes
 $100,000 \times 40 = 4,000,000$ bytes

Overhead = 4 MB

Step 3: Total Data

$150 + 4 = 154$ MB
 $150 + 4 = 154$ MB

Convert to bits

$154 \times 8 = 1232$ Mb
 $154 \times 8 = 1232$ Mb

Transmission time

$\frac{1232}{50} = 24.64$ seconds
 $\frac{1232}{50} = 24.64$ seconds

Transmission Time = 24.64 s

Reliability

- Transport Layer provides segmentation, acknowledgements, retransmission, and flow control.
- Data Link Layer performs framing and error detection using CRC.

2. sliding Window Protocol Question

Total frames = 48

One frame in every window is lost.

Answer

Number of Windows

$$48 \div 6 = \frac{48}{6} = 8$$

Windows = 8

Retransmissions

1 frame lost in every window

$$8 \times 1 = 8$$

Retransmissions = 8

Flow Control

Flow control prevents the sender from overwhelming the receiver, reducing congestion and improving network efficiency.

3. IP Fragmentation Question

Packet size = 12,000 bytes MTU

= 1500 bytes

Header = 20 bytes per fragment

Answer

Payload per fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Fragment

Fragments = 9

Header Overhead

$9 \times 20 = 180 \text{ bytes}$

Header Overhead = 180 bytes

Explanation

The Network Layer fragments packets according to MTU and reassembles them at the destination using identification and fragment offset fields.

4. Sliding Window

Number of windows

$48/6 = 8$

Retransmissions

888

Flow control allows efficient transmission without receiver buffer overflow.

5. Session Layer Synchronization Question

Video size = 600 MB

Checkpoint every 60 MB

Failure after 390 MB

Answer

$$390/60=6.5 \quad 390/60=6.5 \quad 390/60=6.5$$

Completed checkpoints

6 checkpoints

Last checkpoint

$$6 \times 60 = 360 \text{ MB} \quad 6 \times 60 = 360 \text{ MB} \quad 6 \times 60 = 360 \text{ MB}$$

Retransmitted data

$$390 - 360 = 30 \text{ MB} \quad 390 - 360 = 30 \text{ MB} \quad 390 - 360 = 30 \text{ MB}$$

Data Retransmitted = 30 MB

Importance

Synchronization checkpoints allow communication to resume from the last saved point instead of restarting the entire transfer.

6. Presentation Layer Compression Question

Original file = 900 MB

Compression = 40% Encryption

overhead = 5% **Answer**

Compressed Size

$$900 \times 0.60 = 540 \text{ MB} \quad 900 \times 0.60 = 540 \text{ MB} \quad 900 \times 0.60 = 540 \text{ MB}$$

Encrypted Size

$$540 \times 1.05 = 567 \text{ MB} \quad 540 \times 1.05 = 567 \text{ MB} \quad 540 \times 1.05 = 567 \text{ MB}$$

Reduction

$$900 - 567 = 333 \text{ MB} \quad 900 - 567 = 333 \text{ MB} \quad 900 - 567 = 333 \text{ MB}$$

Percentage Reduction

$$\frac{333}{900} \times 100 = 37\% \quad \frac{333}{900} \times 100 = 37\%$$

Final Answers

Compressed file = **540 MB**

Encrypted file = **567 MB**

Reduction = **37%**

Explanation

The Presentation Layer compresses data to reduce transmission time and encrypts it to ensure confidentiality and security.

7. FTP Data Transfer Question

File = 3 GB

Request size = 3 MB

Throughput = 120 Mbps

Answer

Convert

3 GB = 3000 MB

Requests

$$3000/3 = 1000 \quad 3000/3 = 1000 \quad 3000/3 = 1000$$

Requests = 1000

Transmission Time

3000 MB = 24,000 Mb

$24000/120=200$ seconds
 $24000/120=200\text{ seconds}$
 $24000/120=200$ seconds

Transmission Time = 200 seconds

Explanation

The Application Layer provides services like FTP, user authentication, file management, and reliable data transfer.

8. End-to-End

Delay Question

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Propagation = 18 ms

Transmission = 12 ms

Answer

Processing Delay

$4+3+5+2+1=15$ ms
 $4+3+5+2+1=15\text{ ms}$
 $4+3+5+2+1=15$ ms

Total Delay

$15+18+12=45$ ms
 $15+18+12=45\text{ ms}$
 $15+18+12=45$ ms

Total End-to-End Delay = 45 ms

Explanation

Each OSI layer contributes processing time, while transmission and propagation delays determine the total communication delay.

9. Network Efficiency Question

Useful Data = 80 MB

Frame size = 1600 bytes

Overhead = 80 bytes

Answer

Useful payload/frame

$1600 - 80 = 1520$ bytes $1600 - 80 = 1520$ bytes Frames

Required

$80,000,000 / 1520 = 52,632$ frames (approx.) $80,000,000 / 1520 = 52,632$ frames (approx.)

Overhead

$52,632 \times 80 = 4,210,560$ bytes $52,632 \times 80 = 4,210,560$ bytes

≈ 4.21 MB

Efficiency

$\frac{1520}{1600} \times 100 = 95\%$

Explanation

Protocol overhead consumes bandwidth. Lower overhead increases network efficiency and throughput.

10. Hospital IoT Network Question

Original File = 240 MB

Compression = 30%

Segment size = 1400 bytes

Frame overhead = 60 bytes

Answer

Step 1: Compressed File

$240 \times 0.70 = 168$ MB $240 \times 0.70 = 168 \text{ MB}$

Step 2: Convert

168 MB

$168,000,000$ bytes $168,000,000 \text{ bytes}$

Step 3: Segments

$168,000,000 / 1400 = 120,000$ $168,000,000 / 1400 = 120,000$ $168,000,000 / 1400 = 120,000$

Segments = 120,000

Step 4: Protocol Overhead

$120,000 \times 60 = 7,200,000$ bytes $120,000 \times 60 = 7,200,000 \text{ bytes}$ $120,000 \times 60 = 7,200,000$ bytes

= 7.2 MB

Step 5: Final Data

$168 + 7.2 = 175.2$ MB $168 + 7.2 = 175.2 \text{ MB}$ $168 + 7.2 = 175.2$ MB

Final Answers

- **Compressed file size = 168 MB**
- **Segments = 120,000**

- **Protocol overhead = 7.2 MB**
 - **Final transmitted data = 175.2 MB Explanation**
 - **Presentation Layer:** Compresses and secures data.
 - **Transport Layer:** Segments data and ensures reliable delivery.
 - **Data Link Layer:** Adds framing and error detection.
- Physical Layer:** Transmits the final bit stream over the